

4. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

- ☐ This memory address does not map to actual memory cell that corresponds to address number.
- ☐ This memory is the same as physical memory.
- ☐ This memory cannot be accessed by applications at run time.
- ☐ This memory address is always present even after a system is restarted.

Clear Selection

5. What is the postfix expression for the corresponding infix expression?

What is the postfix expression for the corresponding infix expression? $a+b^*c+(d^*a)$

Pick ONE option

☐ abc^*+de^*+

☐ $abc+^*de^*+$

☐ $a+bc^*de+^*$

☐ $abc^*+(de)^*+$

Clear Selection

```
#include <stdio.h>
#define MAX(a,b) ((a) > (b) ? (a) : (b))

int main() {
    int x = 5, y = 10;
    printf(" %d %d %d\n", MAX(x++, y++), x, y);
    return 0;
}
```

Trying to reconnect.

Pick **ONE** option

☐ 11 6 12

☐ 12 6 12

Connected to network.

7. Consider the below java method. What's the output of this method? `public static void function_1(int [] data, int low, int high) { if (low < high) { int temp = data[low]; data[low] = data[high]; data[high] = temp; function_1(data, low + 1, high - 1); } }`

Consider the below java method. What's the output of this method?

```
public static void function_1(int [] data, int low, int high) {  
    if (low < high) {  
        int temp = data[low];  
        data[low] = data[high];  
        data[high] = temp;  
        function_1(data, low + 1, high - 1);  
    }  
}
```

Pick **ONE** option



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8. Consider a database table "orders" with columns: order_id (Primary Key), customer_id, order_date, and total_amount. Which of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Consider a database table "orders" with columns: order_id (Primary Key), customer_id, order_date, and total_amount. Which of the following SQL statements would correctly create a partitioned table based on the order_date column, partitioned by range?

Pick **ONE** option

☐ create table orders partition by range columns (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);

☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2) primary key (order_id);

☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition p2 values less than ('2024-01-01') tablespace ts2);

☐ create table orders partition by range (order_date) (partition p1 values less than ('2023-01-01') tablespace ts1, partition



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14. What is the use of /etc/hosts file in Linux ?

What is the use of /etc/hosts file in Linux ?

Pick **ONE** option

☐

It contains the list of all users.

☐

It contains a list of all files in the system

☐

It contains the time and date information.

☐

It contains the FQDN to IP address mappings.

What does CAP theorem describe with respect to System Design ?

Pick **ONE** option

- ☐ The relationship between CPU and memory usage
- ☐ The trade-offs between consistency, availability, and partition tolerance
- ☐ The process of data encryption
- ☐ The principles of object-oriented programming

Clear Selection

Which criteria are not valid/suitable for Non-Relational Databases?

Which of the following design criteria are not valid/suitable for Non-Relational Databases?

Pick **ONE** option

☐ Your application required super-low latency

☐ Your data are unstructured without any relation between them

☐ You only need to serialize & deserialize data (JSON, XML, YAML)

☐ You need to store a very small amount of data

Clear Selection

15. Which Clause restricts the groups of rows returned to those groups for the specified condition

Which Clause restricts the groups of rows returned to those groups for the specified condition id True

Pick **ONE** option

☐ Where clause

☐ Having Clause

☐ Distinct

☐ Exists

Clear Selection



Q Search



16. What does idempotency mean in REST API design?

What does idempotency mean in REST API design?

Pick **ONE** option

- ☐ The ability of an API to handle multiple requests in parallel
- ☐ The property that a method can be called multiple times without different outcomes
- ☐ The capability of an API to update data
- ☐ The feature of an API that allows it to delete resources

Clear Selection

What is the purpose of ARP (Address Resolution Protocol) in computer networking?

Pick **ONE** option

- ☐ To convert IP addresses to MAC addresses
- ☐ To convert domain names to IP addresses
- ☐ To establish a secure connection between client and server
- ☐ To route packets between networks

Clear Selection

18. In a ring topology network, if one device fails, how does it affect the network?

In a ring topology network, if one device fails, how does it affect the network?

Pick **ONE** option

☐ Only the failed device is affected

☐ All devices in the network are affected

☐ Only the devices between which the failed device exists are affected

☐ The network continues to function normally

Clear Selection

20. In what way the Symmetry Sparse Matrix can be stored efficiently?

In what way the Symmetry Sparse Matrix can be stored efficiently?

Pick **ONE** option

☐ Heap

☐ Binary tree

☐ Hash table

☐ Adjacency List

Clear Selection

Out of the nine ingredients, a chef wants to use exactly five. Given the following statements, how many of the 5 ingredients are known?

The ingredients are apricots, bacon, cake, donuts, eggs, figs, grapes, hazelnuts, and ice cream.

1. If bacon is not used, then grapes, hazelnuts, and ice cream are not used.
2. If apricots are used then bacon and cake are used.
3. If donuts are not used, apricots are not used.
4. Apricots and eggs are used.

Pick **ONE** option

☐ 2

☐ 3

☐ 4

☐ 5

ALL



21. Which of the following is false?

Which of the following is false?

Pick **ONE** option

- ☐ 2-3 tree requires less storage than the BST
- ☐ lookup in 2-3 tree is more efficient than in BST
- ☐ 2-3 tree is shallower than BST
- ☐ 2-3 tree is a balanced tree

Clear Selection

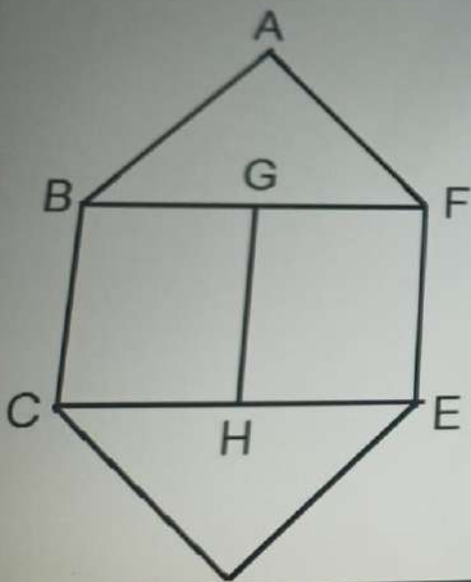
23. Paths in a Graph 2

Answered: 20 / 36

01 hour 10 mins

Bhavani Kondra

There is a network of eight cities (represented by vertices) connected by roads. A traveler must drive on each road segment once but can visit a city multiple times if necessary. Furthermore, the starting and ending points do not have to match. New roads are constructed between C and G, and between H and F. If the journey begins at B, where will it end?



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24. Painted Cube

A cube is painted green on all surfaces and is cut into 1000 identical cubes. How many of the smaller cubes have three faces painted?

Pick **ONE** option

☐ 8

☐ 10

☐ 100

☐ 64

Clear Selection

25. Favorite Color

There are five people: Jack, Matthew, Albert, Peter, and Sebastian.
Each plays one sport out of football, cricket, volleyball, badminton, and squash.
They are from South Korea, the UK, India, China, and Russia, though not necessarily in that order.
Their favorite colors are brown, green, red, black, and yellow, again not necessarily in that order.

The following information is known-

1. Jack's favorite color is red. He is not from India or China.
2. Albert plays football. His favorite color is not yellow, and he is not from the UK.
3. Peter and Sebastian have yellow and green as their favorite colors, not necessarily in that order.
4. Matthew plays badminton. He is from Russia.
5. Peter and Sebastian are from China and UK, not necessarily in that order.
6. Brown is the favorite color of the person from India.
7. Green is the favorite color of the person from China.
8. The person from South Korea plays squash and the person from the UK plays volleyball. One of them has yellow as his favorite color.

Whose favorite color is black?

Pick **ONE** option

☐ Matthew

26. Standing Order 2

There are five people - Peter, Tyson, Richard, Steve and Quinn - standing in a row. Given the following statements, who is in the middle?

1. Peter is next to Quinn and Steve is next to Richard.
2. Steve is not next to Tyson.
3. Tyson is on leftmost position.
4. Richard is on the second position from the right.
5. Peter stands somewhere to the right of both Quinn and Tyson.
6. Peter and Richard are next to each other.

Pick **ONE** option

☐ Peter

☐ Richard

☐ Steve

☐ Quin

29. I don't think we should hire him. He seems like _____.

I don't think we should hire him. He seems like _____.

Pick **ONE** option

☐ A fish out of water

☐ The black sheep of the family

☐ A wolf in sheep's clothing

☐ The apple of my eye

Clear Selection

self-governing country or region

correct option which can be substituted

In the following question, choose the correct option which can be substituted for the given sentence.
A self-governing country or region

Pick **ONE** option

☐ Autonomy

☐ Autocracy

☐ Anarchy

☐ Ethnology

Clear Selection

30. I've been studying for this exam for months, so I'm hoping to _____.

I've been studying for this exam for months, so I'm hoping to _____.

Pick **ONE** option

☐ Hit the ground running

☐ Have a field day

☐ Be a dark horse

☐ Have a lot on my plate

Clear Selection

36. How Many Like Cricket?

In a group of students, 60% like football, 40% like cricket. What percentage of students like only cricket if 12% of the students like neither football nor cricket?

Pick **ONE** option

☐ 28

☐ 14

☐ 56

☐ 42

Clear Selection



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whose favorite color is black?

Pick **ONE** option



Matthew



Albert



Peter



Sebastian

Clear Selection



Se

55. Logic - Sports enthusiasts

In a company, 100 employees are randomly selected. 50 like watching football 40 like watching hockey and the rest like both. Find out the number of employees who like to watch at least one of the games.

Pick **ONE** option

☐ 40☐ 90☐ 80☐ 20

Clear Selection

Read the passage and answer the questions that follow.

While the first industrial revolution, which spanned the years from roughly 1760 to about 1820-1840, laid the groundwork, the second industrial revolution brought about changes truly desired by both the masses and high-profile citizens. The second industrial revolution, as defined by historians, took place between 1870 and 1914. At the time, there was an abrupt rise in urbanization and the development of additional land. This created the need for technological breakthroughs to make living simpler.

The steel industry boomed while iron became almost obsolete. Steel's lower cost and higher strength-to-weight ratio made it simpler and more practical for building projects, industrial machinery, railroads, ships, and many other things. As a result, building bigger structures, creating more mechanical commodities, and traveling and transporting goods farther distances became common.

The development of electricity into wide use was even more significant. Before the arrival of public electricity, candles and gas lamps were used to light houses and factories. Electric lighting altered how people worked and lived, as working hours were no longer restricted to daytime. Beyond a source of illumination, public electricity spawned numerous other inventions. Alexander Graham Bell invented the telephone in 1876. This inspired Guglielmo Marconi to invent the practical radio wave-based wireless telegraph system that could send signals over the Atlantic Ocean in 1901. Thomas Edison perfected the light bulb in 1879, which was reported to last 13 and a half hours in the wake of several unsatisfactory attempts by numerous inventors.

The first electric railroad debuted in Berlin, Germany, in 1879. Likewise, electric streetcars began to replace horse-drawn carriages in major European cities as early as 1880.

The crown of the second industrial revolution was the internal combustion engine. It was first presented in 1878. The gas and the air-driven vehicle was unsuitable for the general public due to safety issues. With the advent of more stable liquid fuels such as petroleum, initial problems were quickly overcome.

Automobiles and airplanes were developed as a result of this engine. None of the following innovations would have been conceivable without it, meaning that the modern means of transportation would be only a dream.

German inventor Karl Benz made the first self-propelled car in 1886. It was made available to the public in 1888. In the same year, Gottlieb



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The German inventor Karl Benz made the first self-propelled car in 1886. It was made available to the public in 1888. In the same year, Gottlieb Daimler developed a lighter-weight engine, and in 1896, Henry Ford fully commercialized the use of automobiles.

Years later, the Wright brothers invented the first practical aircraft, overcoming the difficult technical issues that had impeded mechanical flight for centuries. Before their invention, steerable or dirigible hydrogen-powered balloons were the only method of air travel available to man. Since then, the world has experienced several industrial revolutions that have paved the way for the development of electronics, telecommunications, and computers and opened the door for space exploration, research, and biotechnology. Still, the second revolution is still remembered as the era of iconic inventions that spawned technology use today.

The word "spawned" in the passage is farthest in meaning from



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As the 18th century came to a close, France and Britain were in a drawn-out war to advance their empires. Due mainly to the cost of war, France was on the verge of bankruptcy. Meanwhile, French King Louis XVI and his queen, Mary Antoinette, lived an extravagant lifestyle. Despite the fact that he ruled France during the height of the Enlightenment Era, Louis XVI was unable to establish himself as an enlightened absolutist. His efforts to allow citizens more opportunities along with some proposed societal ideals led to increased bureaucracies and even more expenditures. The royal treasury was almost empty, necessitating additional taxes. At the same time, there were years of poor harvests due to drought. Spiking bread prices led to unrest among the non-aristocrat classes. This, coupled with increased taxes, caused resentment among the lower classes. Even some in the noble classes became disenchanted with the monarchy as their own access to a lavish lifestyle was curtailed. As the war continued, citizens became agitated and began to speak out against the weakened monarchy. The excesses of the monarchy showed citizens that any spoils of war were not likely to be shared with them. Quickly, their unrest changed from passive to active, and common people began plans to depose the monarchy. The nation was overtaken by a wave of mass panic and revolutionary passion. Peasants robbed and set fire to the homes of tax collectors, landlords, and members of the aristocratic class. Out of concern for his family's safety and disgust with the ongoing revolution in the country, in June 1792, the king fled Paris for the Austrian frontier. During the journey, he was recognized, captured, and returned to Paris. An irate public demanded his imprisonment in August 1792, and the monarchy was overthrown in September. Soon after, evidence of the king's counter-revolutionary plots with other countries was discovered, and he was charged with treason. He was found guilty in January 1793, and given the death penalty. He stubbornly approached the guillotine on January 21 and was put to death. Nine months later, queen Marie-Antoinette was executed for the same reason. Ultimately, the monarchy's end resulted from the royals' penchant for abusing power. Decades of waste and obscene indulgences ended in a brutal, extreme manner by a people no longer willing to be yoked. Perhaps the monarchy would still exist if they had paid more attention to the concerns of the populace.

According to the passage, which of the following statements is incorrect?



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subsequently, the word "spattered" is used to describe the way the water droplets are scattered, and the word "spattered" is used to describe the way the water droplets are scattered.

The word "spattered" in the passage is closest in meaning to

select ONE option

☐ fostered

☐ begat

☐ generated

☐ squelched

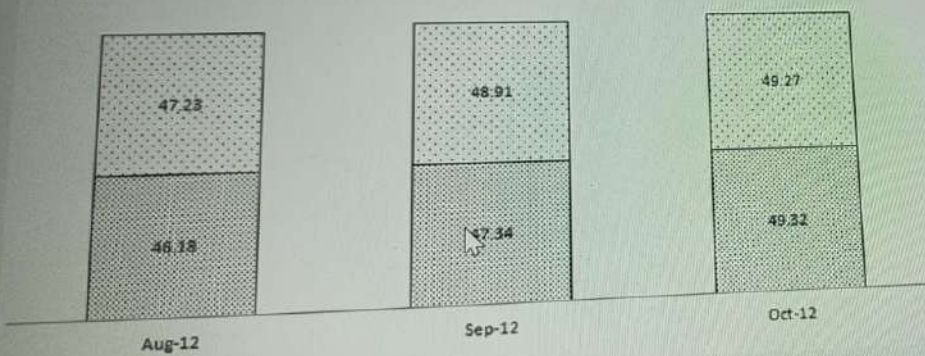
Clear selection

35. Interpret Charts

The following graph shows the number of people (in millions) using Hington and Euphore brand phones in France from August to October 2012. Only these two brands of phones exist in the market.

Number of customers (in millions) using each brand

■ Hington ■ Euphore



Customers in Millions Using Each Phone



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telecommunications, and computers and opened the door for space exploration, research, and b
remembered as the era of iconic inventions that spawned technology use today.

The word "spawned" in the passage is farthest in meaning from

Pick **ONE** option

☐ fostered

☐ begat

☐ generated

☐ squelched

Clear Selection

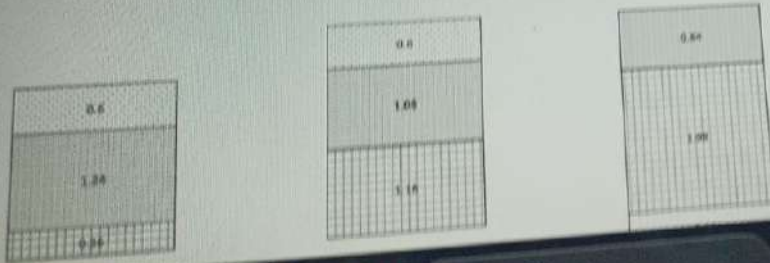
Customers in Millions Using Each Phone

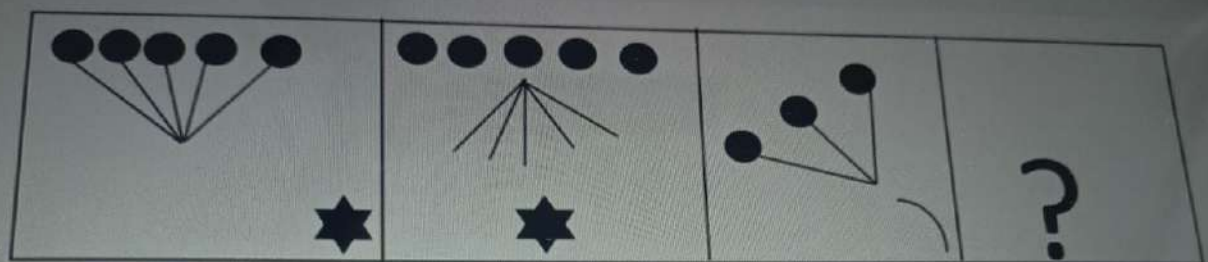
	Hington	Euphore
Aug-12	46.18	47.28
Sep-12	47.34	48.91
Oct-12	49.32	49.27

The following chart shows the addition of customers (in millions) of Hington and Euphore in each of the three months. There are two versions of the Euphore brand phones, Version-1, and Version-2.

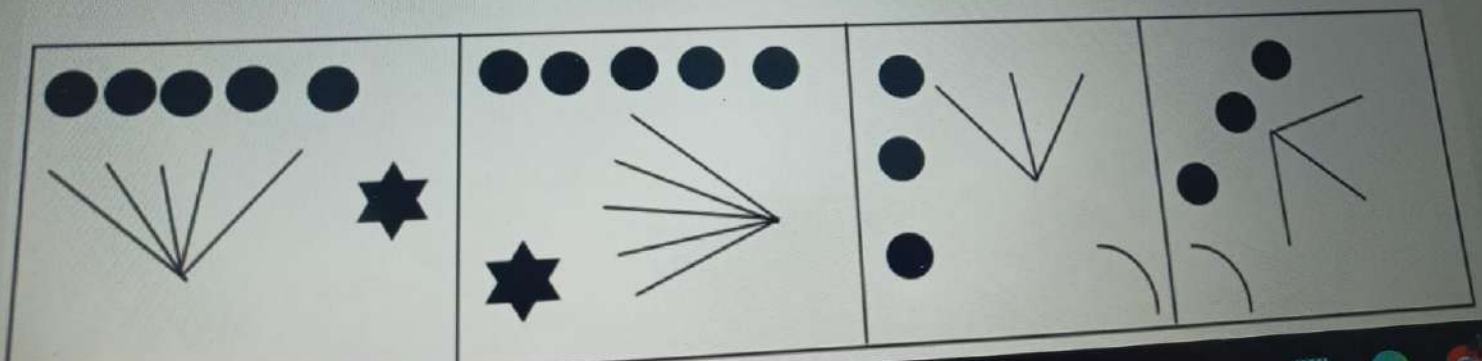
Addition of customers (in millions)

☐ Hington
 ☐ Version-1
 ☐ Version-2





Answer Fig:



left

1. Get the Groups

As new students begin to arrive at college, each receives a unique ID number, 1 to n . Initially, the students do not know one another, and each has a different circle of friends. As the semester progresses, other groups of friends begin to form randomly.

There will be three arrays, each aligned by an index. The first array will contain a *queryType* which will be either *Friend* or *Total*. The next two arrays, *students1* and *students2*, will each contain a student ID. If the query type is *Friend*, the two students become friends. If the query type is *Total*, report the sum of the sizes of each group of friends for the two students.

Example

$n = 4$

queryType = ['Friend', 'Friend', 'Total']

students1 = [1, 2, 1]

Language C++20

Autocomplete Error (Please refresh page)

```

1 > #include <bits/stdc++.h>
9
10 /*
11  * Complete the 'getThe
12  *
13  * The function is exp
14  * The function accep
15  * 1. INTEGER n
16  * 2. STRING_ARRAY
17  * 3. INTEGER_ARRA
18  * 4. INTEGER_ARRA
19  */
20
21 vector<int> getThe
vector<int> studen
22
23 }
24
25 > int main() ...

```

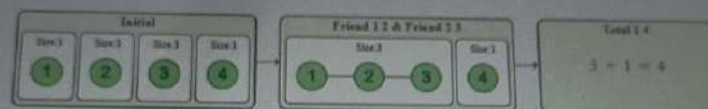
Example

$n = 4$

queryType = ['Friend', 'Friend', 'Total']

student1 = [1, 2, 1]

student2 = [2, 3, 4]



The queries are assembled, aligned by index:

Index	queryType	student1	student2
0	Friend	1	2
1	Friend	2	3
2	Total	1	4

Students will start as discrete groups $\{1\}$, $\{2\}$, $\{3\}$ and $\{4\}$. Students 1 and 2 become friends with the first query, as well as students 2 and 3 in the second. The new groups are $\{1, 2\}$, $\{2, 3\}$ and $\{4\}$ which simplifies to $\{1, 2, 3\}$ and $\{4\}$. In the third query, the number of friends for student 1 = 3 and student 4 = 1 for a Total = 4. Notice that student 3 is indirectly part of the circle of friends of student 1 because of student 2.

Language C++20

Autocomplete Error (Please refer...

```

1 > #include <bits/stdc++.h>
9
10 /*
11  * Complete the function
12  *
13  * The function
14  * The function
15  * 1. INTEGER
16  * 2. STRING_A
17  * 3. INTEGER
18  * 4. INTEGER
19 */
20
21 vector<int> g
vector<int> s
22 I
23 }
24
25 > int main()

```

```
1 > #include <bits/stdc++.h>
9
10 *
11 * Complete the 'getTheGroups' function below,
12 *
13 * The function is expected to return an INTEGER_ARRAY.
14 * The function accepts following parameters:
15 * 1. INTEGER n
16 * 2. STRING_ARRAY queryType
17 * 3. INTEGER_ARRAY students1
18 * 4. INTEGER_ARRAY students2
19 */
20
21 vector<int> getTheGroups(int n, vector<string> queryType,
22 vector<int> students1, vector<int> students2) {
23 }
24
25 > int main() ...
```


- $1 \leq \text{students1}[i], \text{students2}[i] \leq n$
- $\text{queryTypes}[i]$ are in the set $\{\text{'Friend'}, \text{'Total'}\}$

▼ Input Format for Custom Testing

Input from stdin will be processed and passed to the function as follows:

The first line contains an integer n , the number of students.

The next line contains an integer q , the number of queries.

Each of the next q lines contains a string $\text{queryType}[i]$ where $1 \leq i \leq q$.

The next line contains an integer q , the number of queries.

Each of the next q lines contains a string $\text{students1}[i]$ where $1 \leq i \leq q$.

The next line contains an integer q , the number of queries.

Each of the next q lines contains a string $\text{students2}[i]$ where $1 \leq i \leq q$.

▼ Sample Case 0

Sample Input 0



Q Search